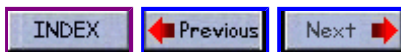


C Programming



File handling using open(), read(), write() and close()

The previous examples of file handling deal with File Control Blocks (FCB). Under MSDOS v3.x (or greater) and UNIX systems, file handling is often done using handles, rather than file control blocks.

Writing programs using handles ensures portability of source code between different operating systems. Using handles allows the programmer to treat the file as a stream of characters.

open()

```
#include <fcntl.h>
int open( char *filename, int access, int permission );
```

The available access modes are

<code>O_RDONLY</code>	<code>O_WRONLY</code>	<code>O_RDWR</code>
<code>O_APPEND</code>	<code>O_BINARY</code>	<code>O_TEXT</code>

The permissions are

```
S_IWRITE          S_IRREAD S_IWRITE | S_IRREAD
```

The `open()` function returns an integer value, which is used to refer to the file. If unsuccessful, it returns -1, and sets the global variable `errno` to indicate the error type.

read()

```
#include <fcntl.h>
int read( int handle, void *buffer, int nbyte );
```

The `read()` function attempts to read `nbytes` from the file associated with `handle`, and places the characters read into `buffer`. If the file is opened using `O_TEXT`, it removes carriage returns and detects the end of the file.

The function returns the number of bytes read. On end-of-file, 0 is returned, on error it returns -1, setting `errno` to indicate the type of error that occurred.

write()

```
#include <fcntl.h>
int write( int handle, void *buffer, int nbyte );
```

The *write()* function attempts to write *nbytes* from *buffer* to the file associated with *handle*. On text files, it expands each LF to a CR/LF.

The function returns the number of bytes written to the file. A return value of -1 indicates an error, with *errno* set appropriately.

close()

```
#include <fcntl.h>
int close( int handle );
```

The *close()* function closes the file associated with *handle*. The function returns 0 if successful, -1 to indicate an error, with *errno* set appropriately.

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